

CHM111 Practice Exam 3 on chapter(s): 5, 6

1. $1.3\% \therefore \Delta x \Delta p \geq \frac{h}{4\pi}$ and $\Delta p = m\Delta v$
2. $330 \text{ zJ} \therefore \text{"C"} \text{ is the level with the third-lowest energy, AKA the second excited state.}$
3. $E \rightarrow A \therefore \text{this transition corresponds to the largest energy difference, resulting in emission of a photon with the shortest wavelength and highest frequency.}$
4. $6 \therefore \text{the number of lines appearing in the spectrum is equal to the number of transitions. Let } n \text{ be the number of excited states; the number of transitions is } n(n+1)/2.$
5. $2.16 \times 10^{-6} \text{ m} \therefore \Delta E = \left(\frac{-R_y}{n_2^2}\right) - \left(\frac{-R_y}{n_1^2}\right) \text{ and } \Delta E = \frac{hc}{\lambda}$
6. Yes $\therefore \text{the de Broglie wavelength } \lambda = \frac{h}{mv} \text{ is larger than the width of the object.}$
7. $4.774 \times 10^{-7} \text{ m}$
8. $-1.82 \times 10^5 \text{ J}$
9. $F \rightarrow A \therefore \text{because this transition corresponds to the largest drop in energy levels, resulting in emission of a photon with the highest frequency (and shortest wavelength).}$
10. 56 kJ
11. 0.077 J/g/K
12. $4p: n = 4, l = 1, N_e = 6 \dots 2s: n = 2, l = 0, N_e = 2 \dots 2p: n = 2, l = 1, N_e = 6$
13. $A \rightarrow F \therefore \text{because this transition corresponds to the largest energy difference, resulting in absorption of a photon with the highest frequency (and shortest wavelength).}$
14. (shortest wavelength) gamma rays < blue light < yellow light < red light (longest wavelength)
15. 3.317 m
16. $45 \text{ zJ} \therefore \text{the energy of the emitted photon is equal to the difference in energy levels. } 135 - 90 = 45$
17. $2s, 2p, 4s, 3d, 4p$
18. $A \rightarrow E \therefore \text{this transition corresponds to the largest energy difference, resulting in absorption of a photon with the shortest wavelength and highest frequency.}$
19. 540 kJ
20. (lowest frequency) microwaves < red light < orange light < gamma rays (highest frequency)
21. $B \rightarrow C \therefore \text{because this transition corresponds to the smallest energy difference, resulting in absorption of a photon with the lowest frequency (and longest wavelength).}$
22. $D \rightarrow C \therefore \text{because this transition corresponds to the smallest drop in energy levels, resulting in emission of a photon with the lowest frequency (and longest wavelength).}$
23. $21 \therefore \text{the number of lines appearing in the spectrum is equal to the number of transitions. Let } n \text{ be the number of excited states; the number of transitions is } n(n+1)/2.$
24. $A \rightarrow B \therefore \text{this transition corresponds to the smallest energy difference, resulting in absorption of a photon with the longest wavelength and lowest frequency.}$

25. $D \rightarrow C$ ∴ this transition corresponds to the smallest energy difference, resulting in emission of a photon with the longest wavelength and lowest frequency.
26. 30 zJ ∴ "A" is the level with the lowest energy, AKA the ground state.
27. 14 J
28. $6.62 \times 10^{-6} \text{ m}$ ∴ the energy of the emitted photon is equal to the difference in energy levels and $E = hc/\lambda$
29. 600 zJ ∴ "B" is the level with the second-lowest energy, AKA the first excited state.
30. -284 kJ. Since $\Delta H < 0$, the reaction is exothermic.
31. (lowest energy) radio waves < orange light < blue light < x-rays (highest energy)
32. Absorbed ∴ a photon is absorbed when an electron transitions to a higher energy level